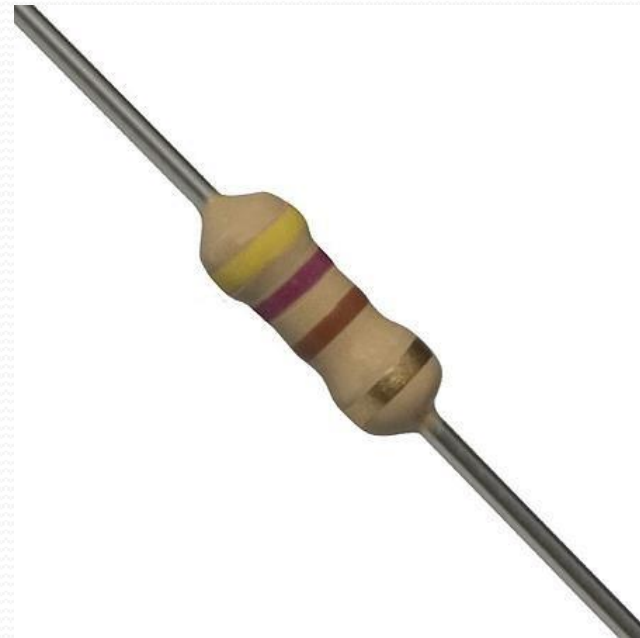


Electronics Engineering

Basic Electronic Components & Measuring Instruments

Resistors

- ❖ A resistor impedes the flow of electricity through a circuit.
 - Resistors have a set value.
 - Since **voltage**, **current** and **resistance** are related through **Ohm's law**, resistors are a good way to control voltage and current.



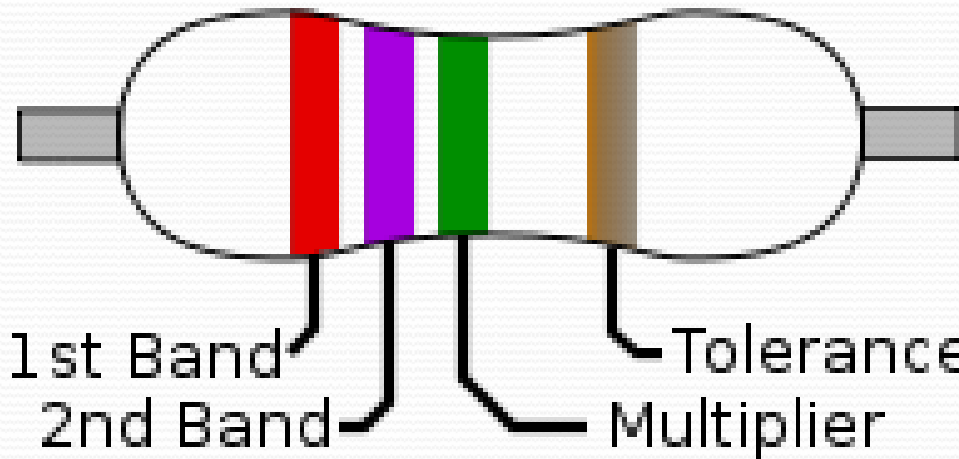
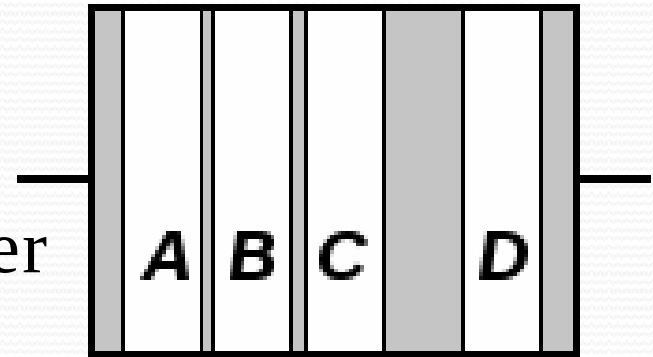
Resistor color codes

1st band = 1st number

2nd band = 2nd number

3rd band = # of zeros / multiplier

4th band = tolerance



Color code

B B R O Y of G B has a V G W

0 1 2 3 4 5 6 7 8 9

❖Tolerance: Gold = within 5%

Black: 0

Brown: 1

Red: 2

Orange: 3

Yellow: 4

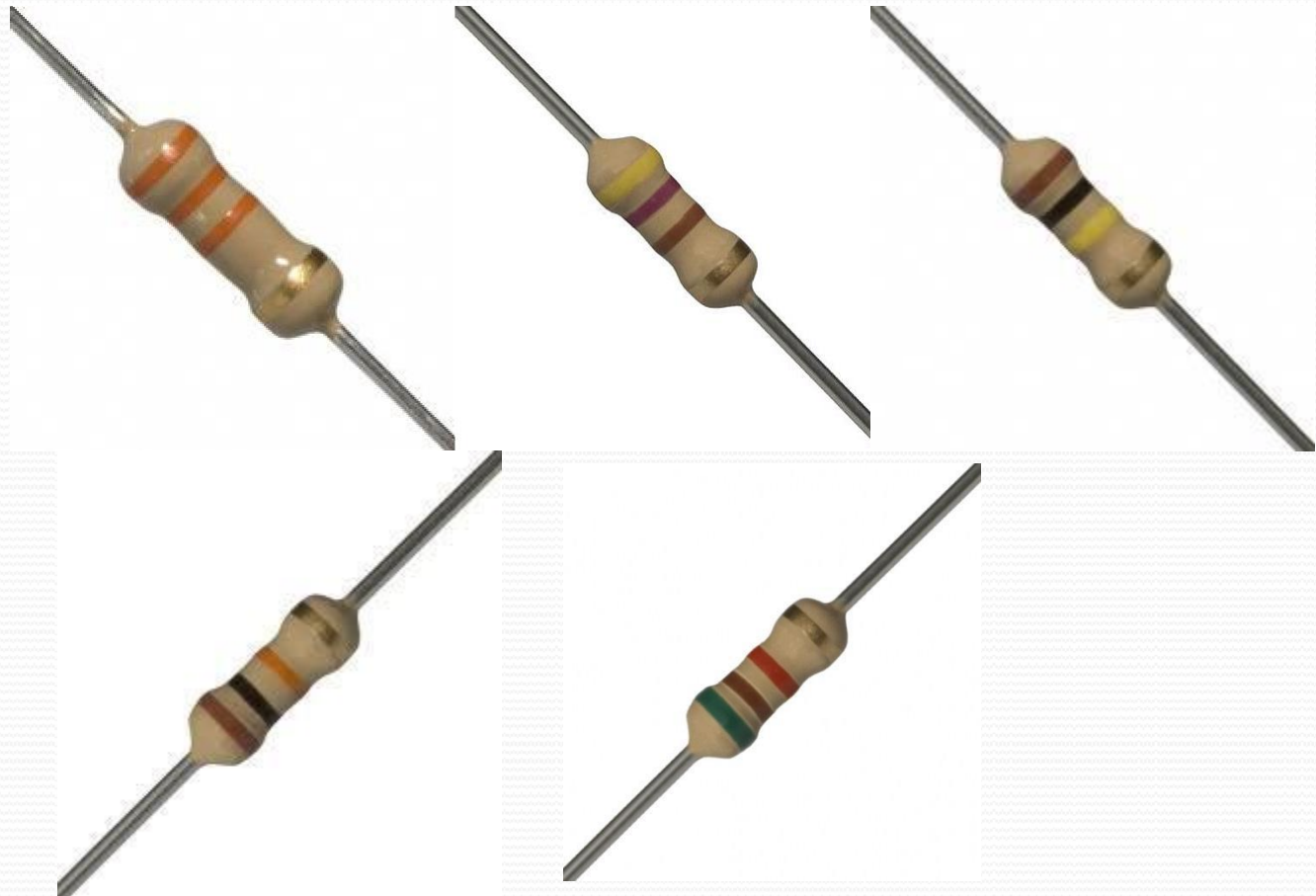
Green: 5

Blue: 6

Violet: 7

Gray: 8

White: 9



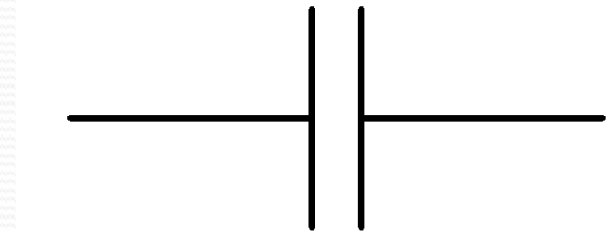
Capacitors

Capacitor stores electrical energy.

- Capacitance is measured in Farads. The small capacitors usually used in electronics are often measured in microfarads and nanofarads.



<u>Value in μF</u>	<u>Value in pF</u>	<u>Capacitor Marking</u>
0.000001 μF	1pF	
0.00001 μF	10pF	
0.0001 μF	100pF	101
0.001 μF	1000pF	102
0.01 μF	10000pF	103
0.1 μF	100000pF	104

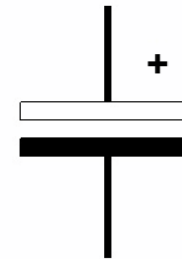
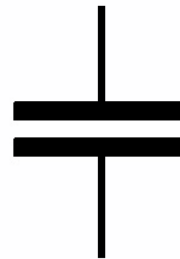
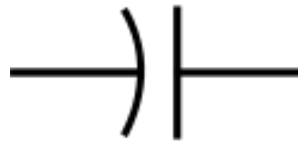
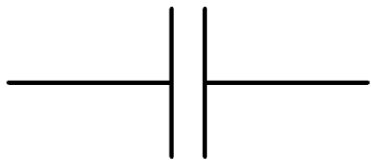


Symbol

Polarity of capacitors

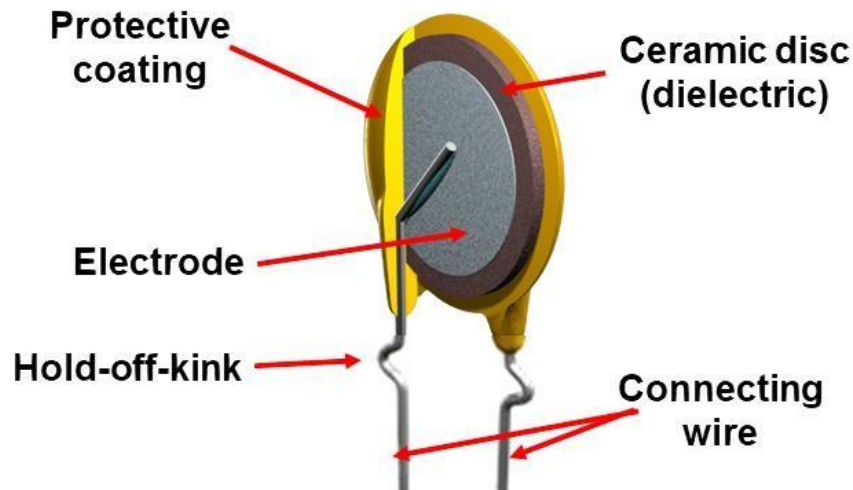
Some capacitors are polarized. Note the different length terminals on one of the capacitors.

- ❖ The shorter terminal goes on the negative side.
- ❖ The stripe is on the negative terminal side of the capacitor.
- ❖ The board is marked for positive or negative.



Applications of capacitors

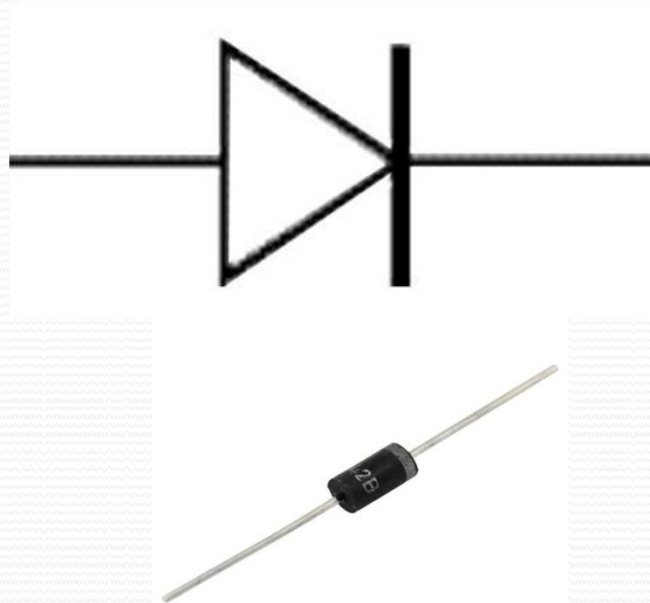
- ❖ Capacitors can **smooth out a signal** – eliminate the ripples or spikes in DC voltage.
 - The capacitor can absorb the peaks and fill in the valleys of a rippled signal.



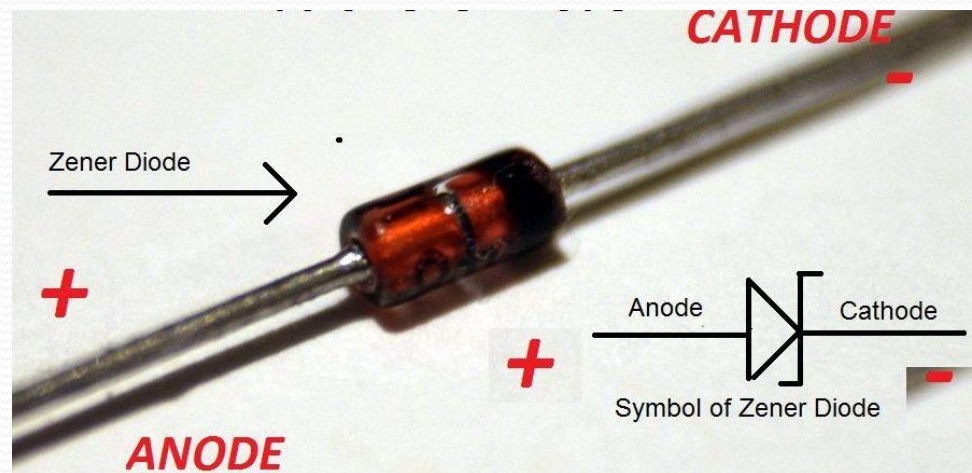
Diode

- ❖ A diode is a **one way valve** (or gate) for electricity. It is a component with an asymmetrical transfer characteristic. A diode has low (ideally **zero**) resistance in one direction, and high (ideally **infinite**) resistance in the other direction.

Rectifier diode



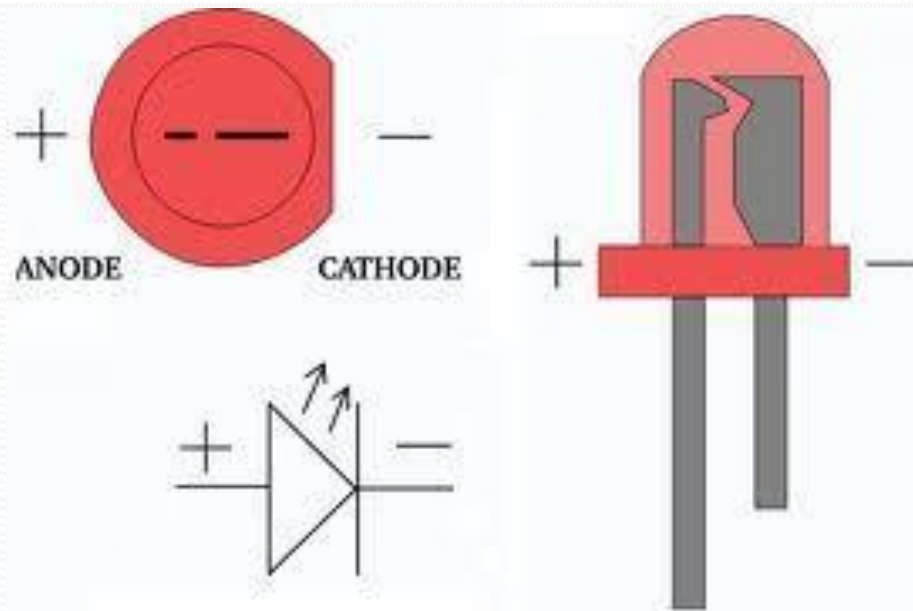
Zener diode



Diodes will protect your electronics.

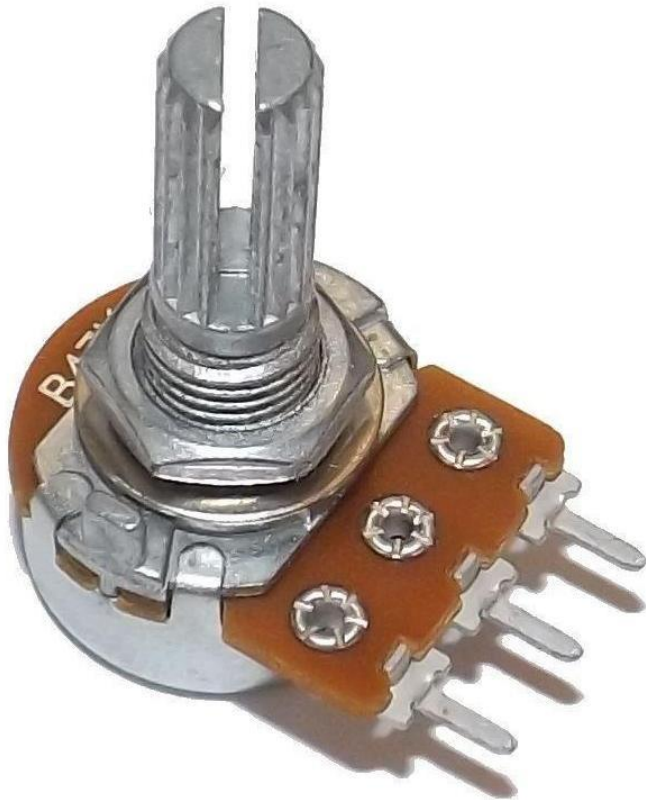
Light emitting diode(LED)

- ❖ A light emitting diode (LED) is a semiconductor light source. When electricity is passing through the diode, it emits light.

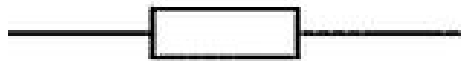


Variable resistor / Potentiometer

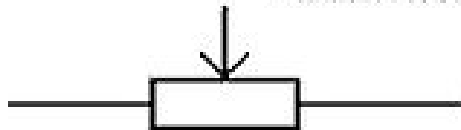
- ❖ A potentiometer is a variable resistor. As you manually turn a dial, the resistance changes.



Fixed value resistor

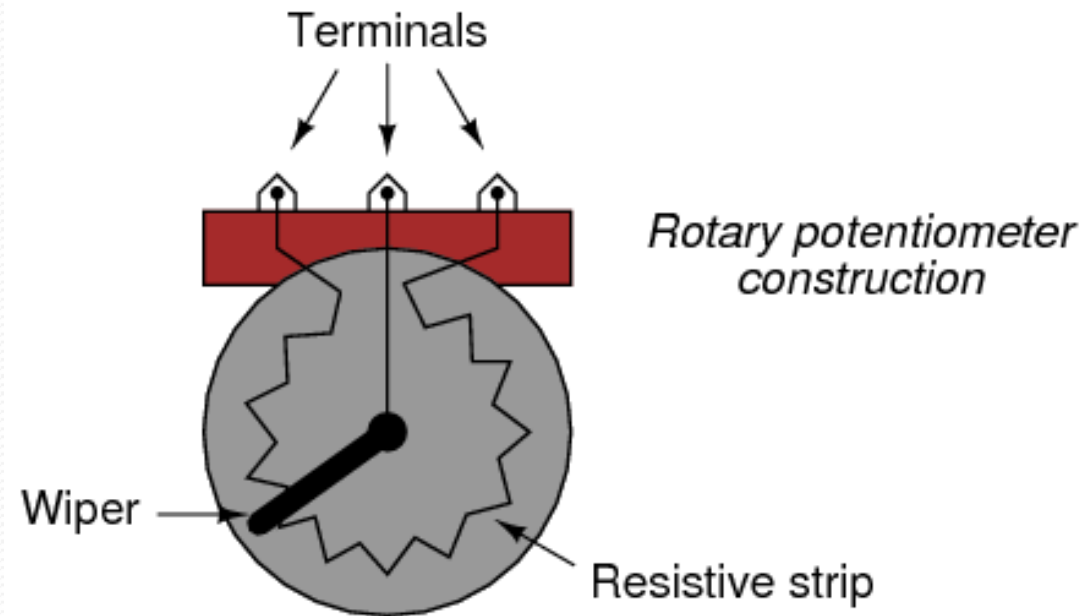


Variable resistor - Potentiometer

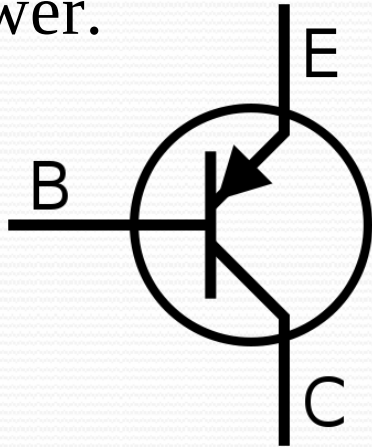


How a variable resistor works

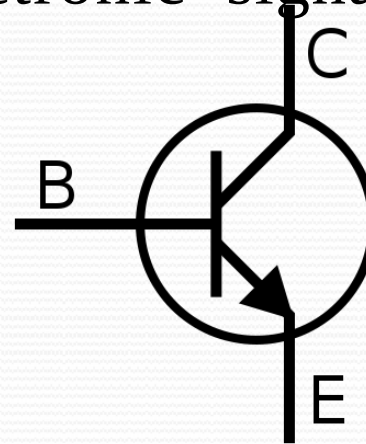
- ❖ As the dial/wiper turns, electricity must go through more or less of the resistive strip.
- ❖ So as you turn the dial/wiper, you get a change in voltage



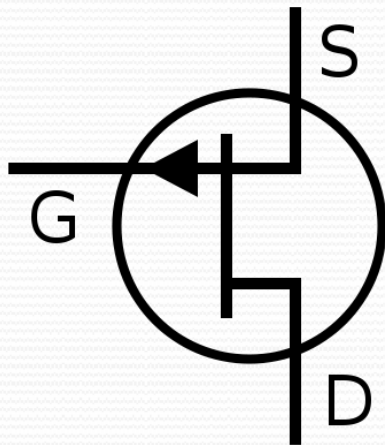
Transistors: Transistor is a semiconductor device used to amplify and switch electronic signals and electrical power.



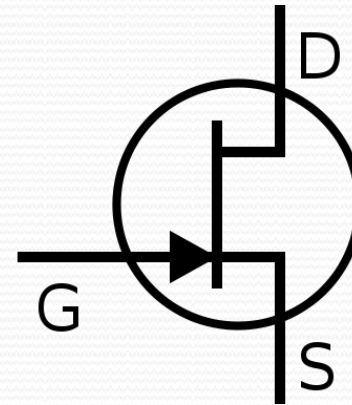
BJT PNP



BJT NPN



P-channel FET



N-channel FET

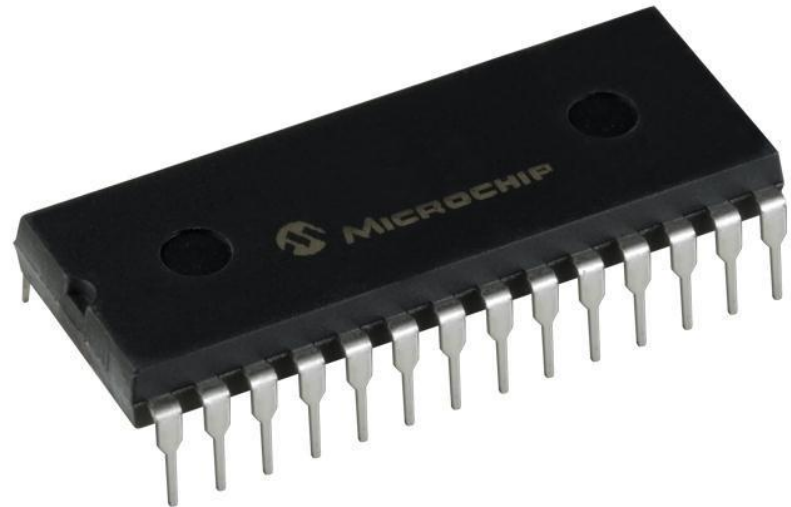
Transistors



- ❖ Metal body is acting as collector for the BJT in the middle
 - To dissipate heat – power transistor
 - Main current flow and hence heat is produced in collector

Integrated Circuit (IC)

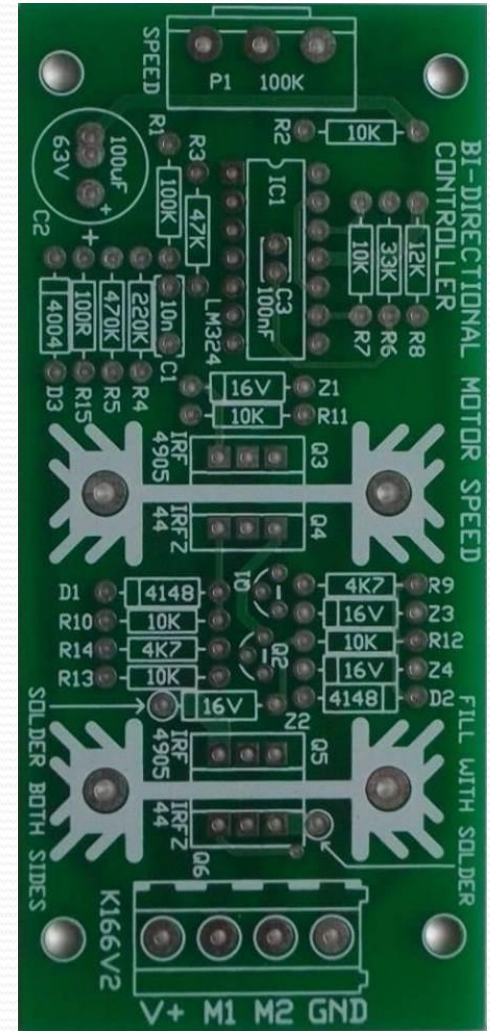
- ❖ An **integrated circuit** (IC) is a set of transistors that is the controller or '**brain**' of an electronic circuit.
 - An input is received, an output is sent out.



Modern microprocessor ICs can have billions of transistors per square inch!

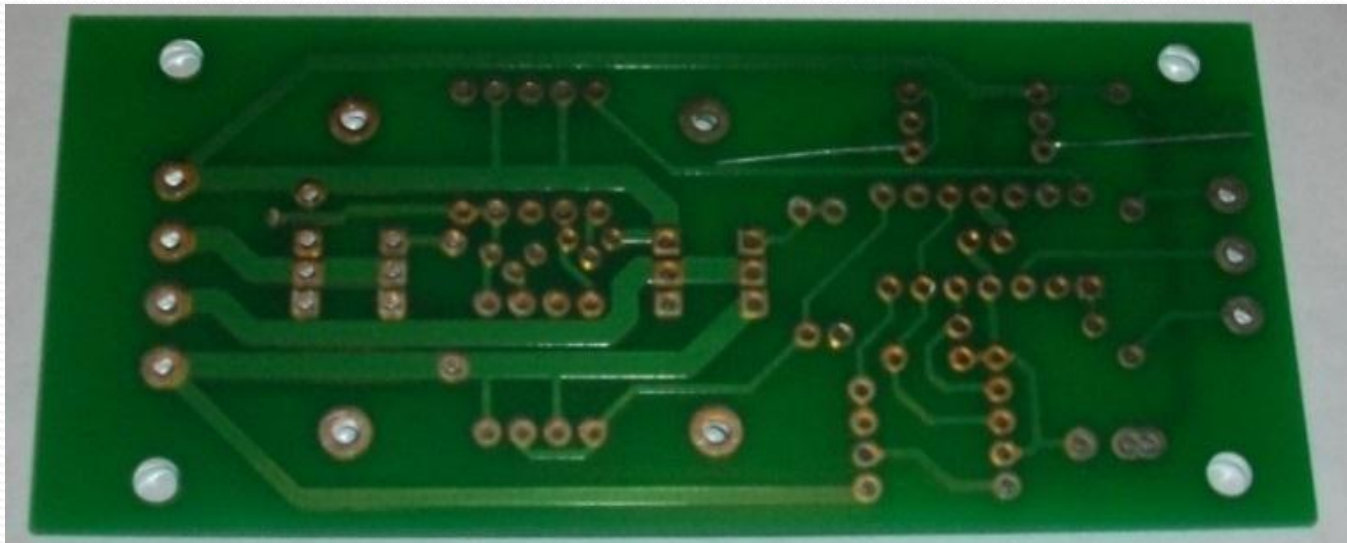
Printed Circuit Board (PCB)

- ❖ Components are attached to a printed circuit board.
- ❖ The '**front**' side of the board will have printed component information, such as resistor, diode type and polarity, etc.
- ❖ Holes go all the way through the board from one side to the other. Through-hole soldering is needed to connect components to the board.



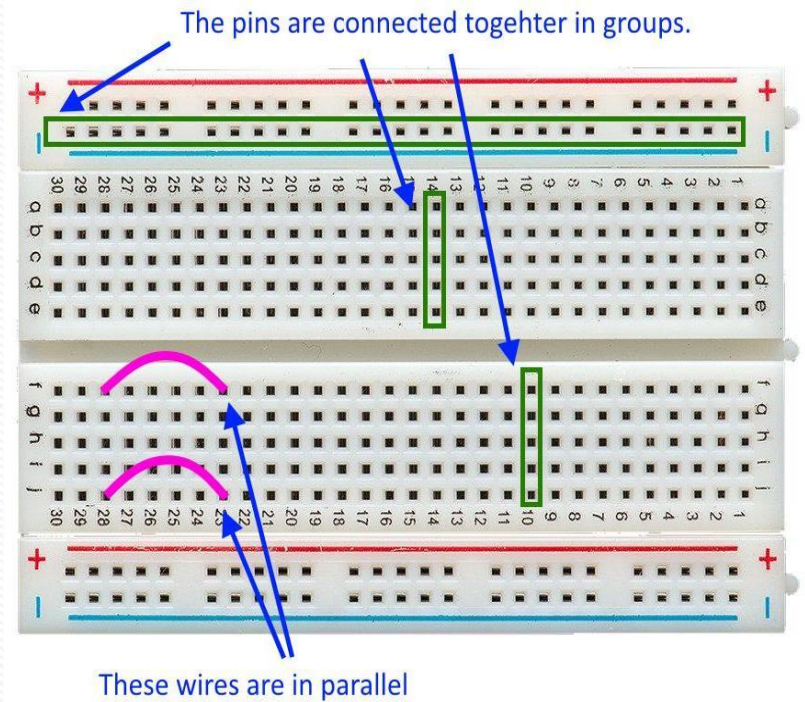
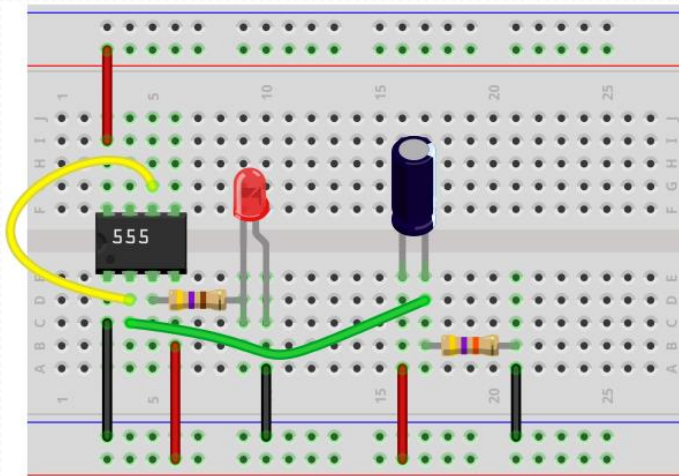
Back of Circuit Board

- ❖ The '**back**' side of the board has lines indicating connections between components. The lines on the back are similar to wires.
- ❖ Thicker lines denote more current (electrons) moving through.

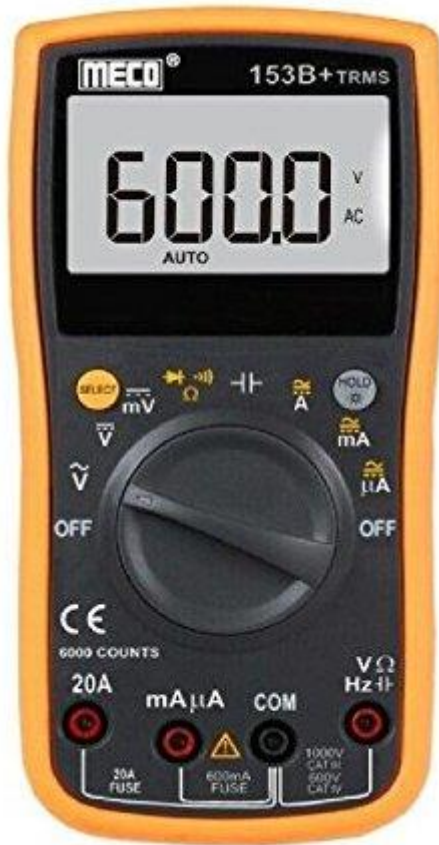


Bread Board

- ❖ To make circuit for testing purpose
- ❖ Some horizontally and some vertically shorted lines are there.

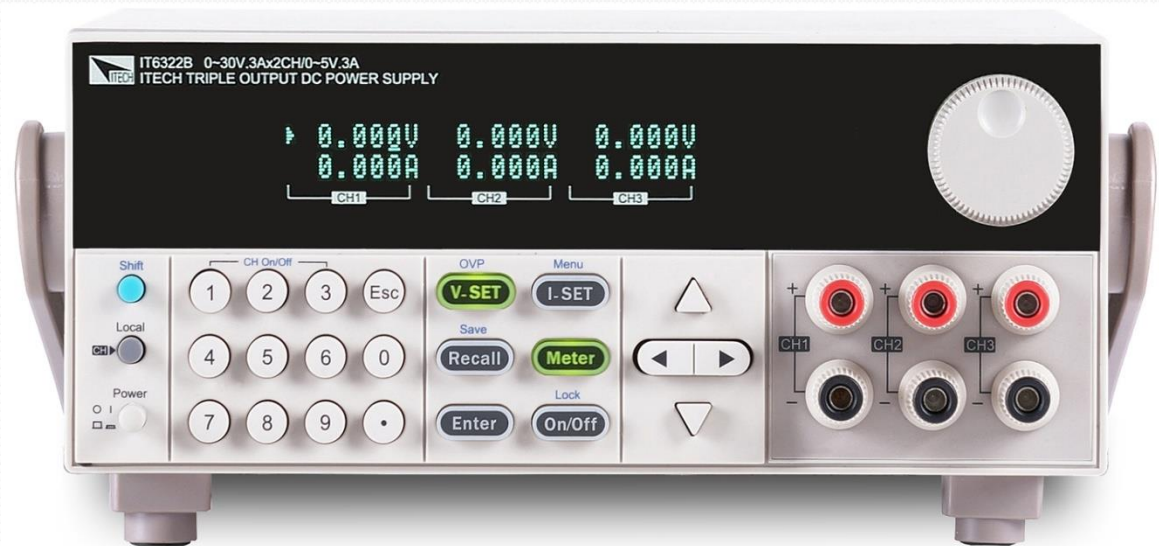


Digital Multimeter



DC Power Supply

❖ Generation of DC supply voltage, like $\pm 5V$, $\pm 10V$, $\pm 30V$ etc.

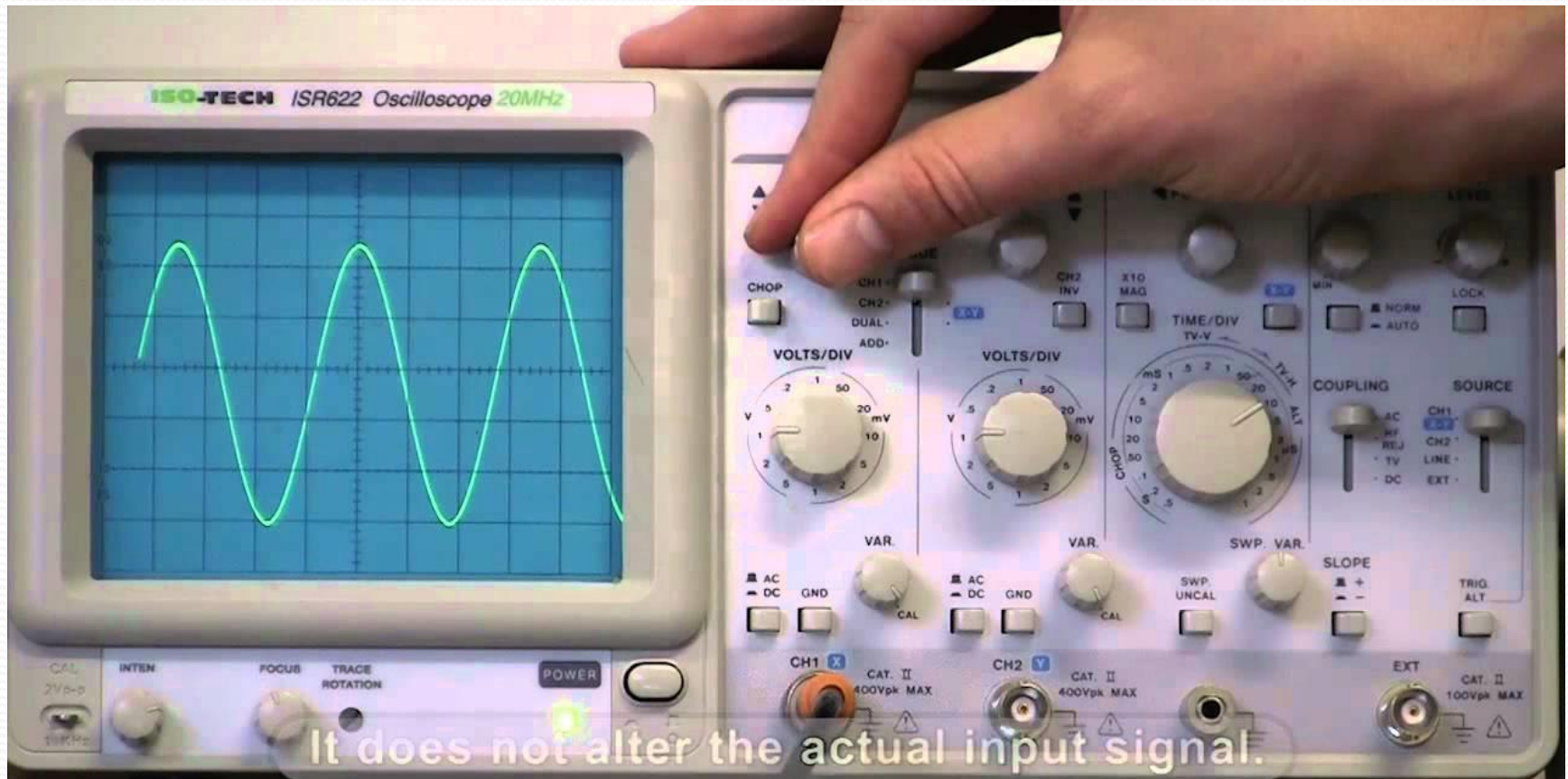


Function Generator

❖ Generates function like sine wave, triangular wave, square wave, pulse etc.



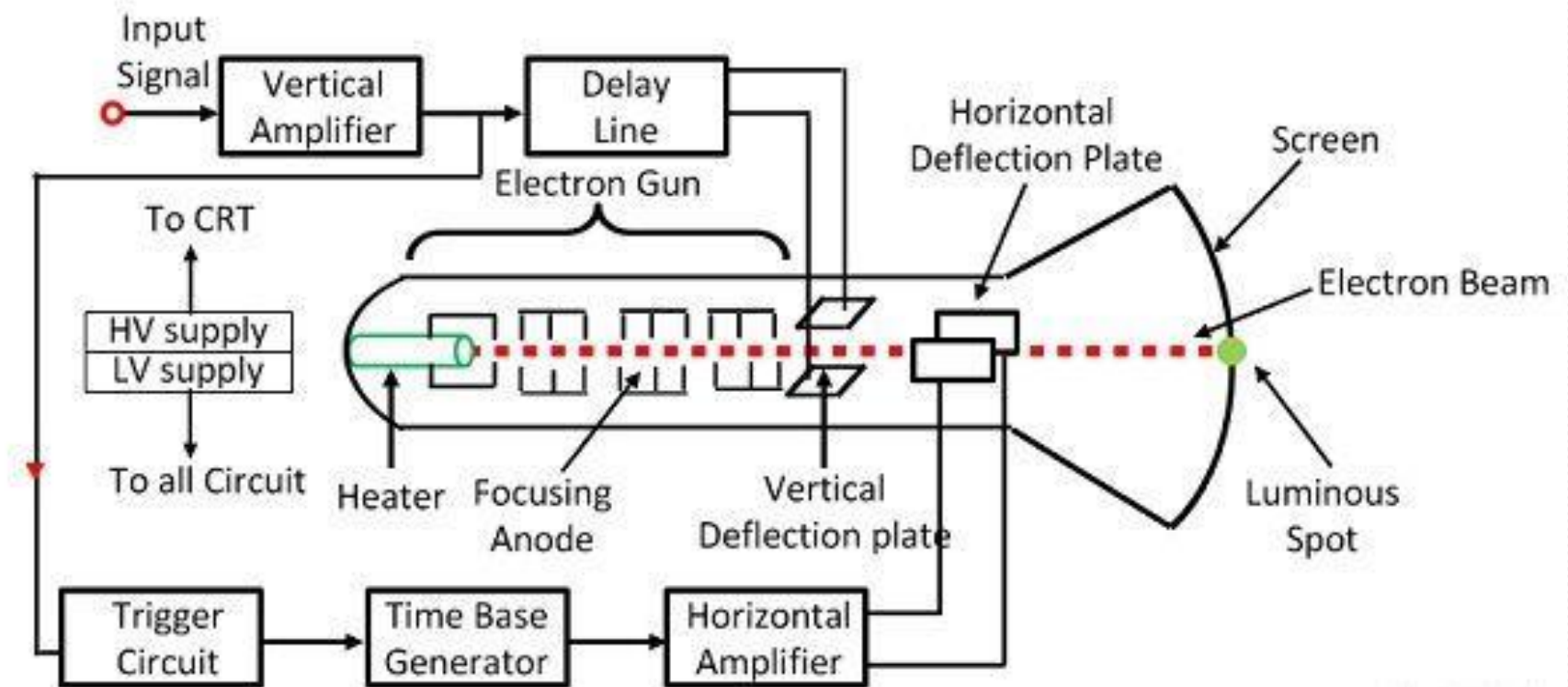
Cathode Ray Oscilloscope (CRO)



- ❖ Time domain display of signals
- ❖ Measurement of voltage, current and frequency, phase of a signal

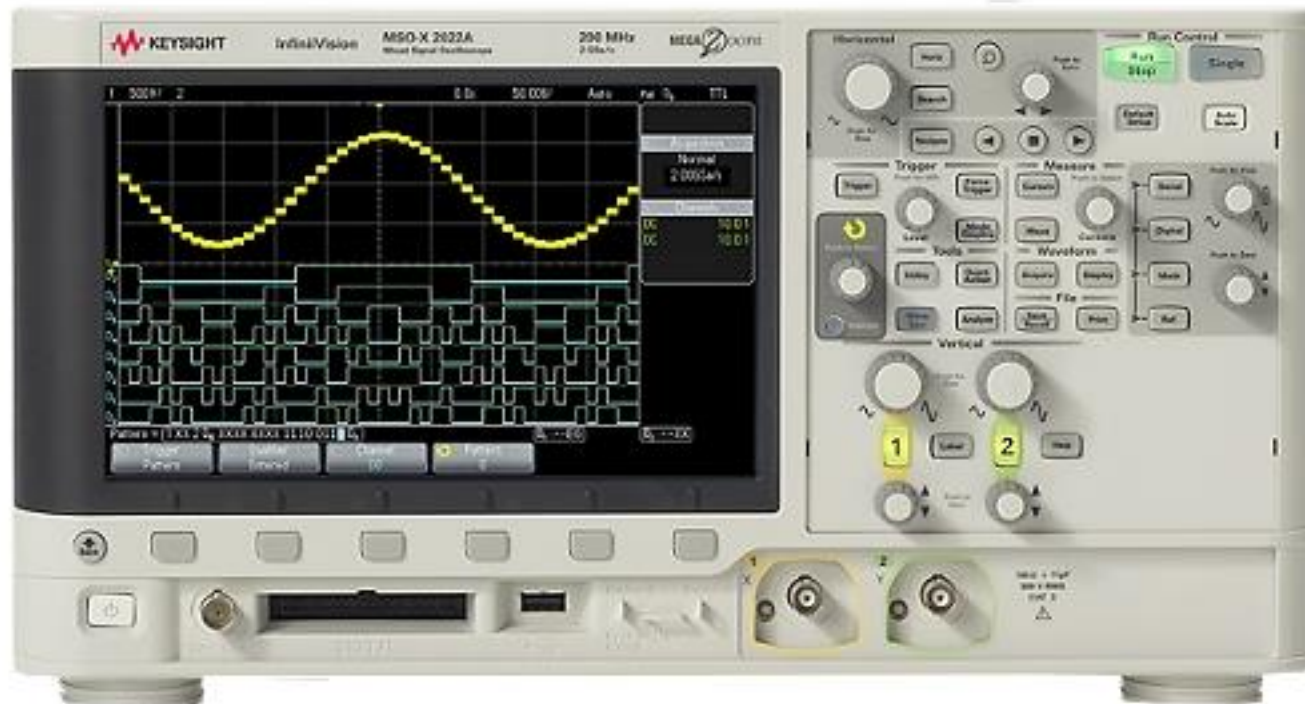
How ??

Cathode Ray Oscilloscope (CRO)



Digital Storage Oscilloscope (CRO)

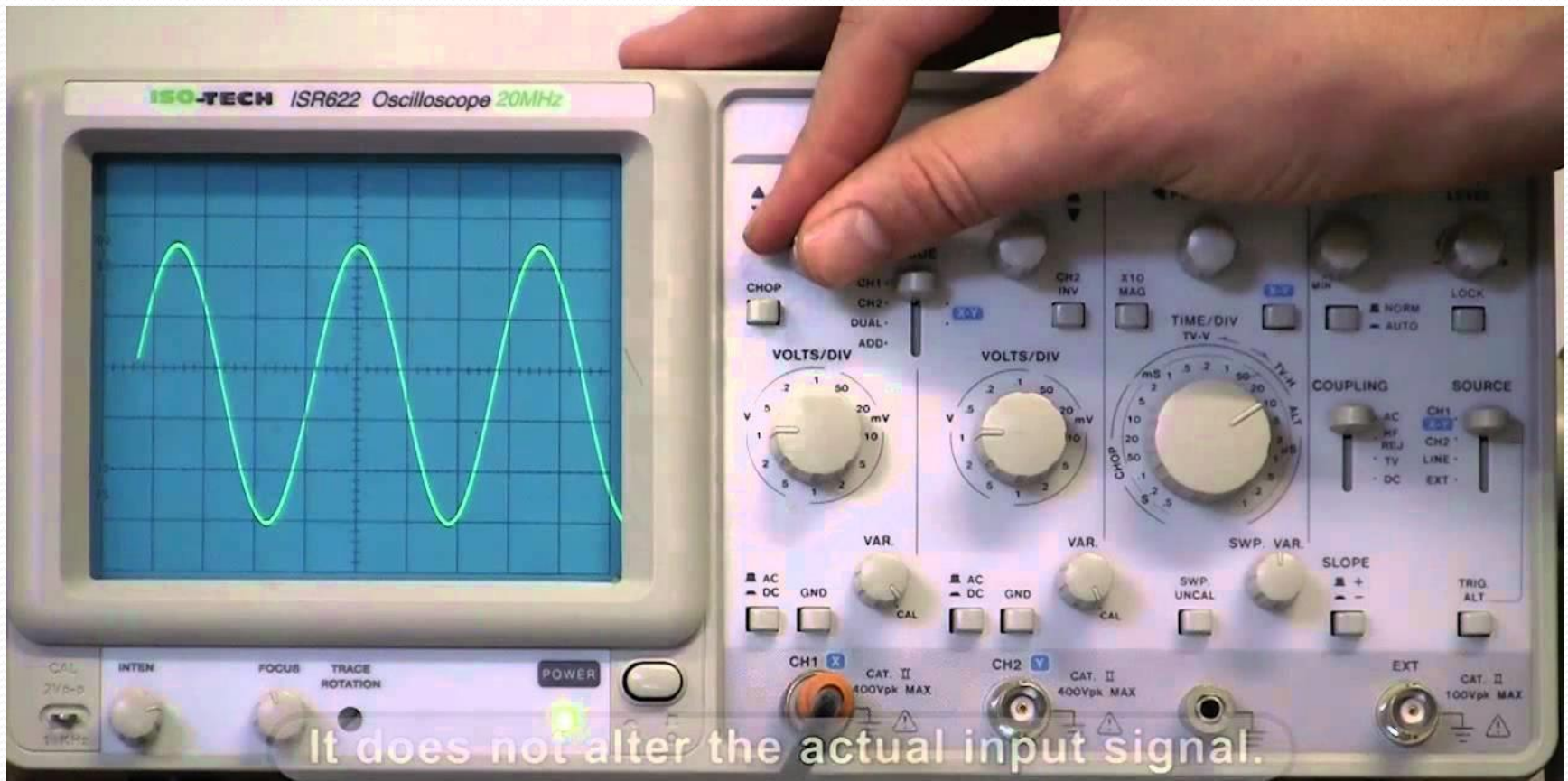
- ❖ To display and measure signals, both analog and digital
 - ✓ Measurement of voltage, frequency, phase, time period etc.
- ❖ Data Storage for live tracking



How to Measure??

Voltage, Frequency, Phase of a signal using Oscilloscope

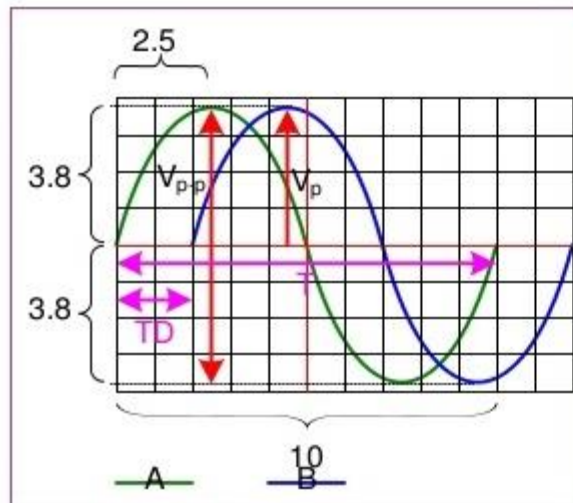
- ❖ Volt/div, time/div ??
- ❖ For current: Voltage across a known resistance
- ❖ Phase: Lissajous figures



How to Measure??

Voltage, Frequency, Phase of a signal using Oscilloscope

- ❖ Volt/div, time/div ??
- ❖ For current: Voltage across a known resistance
- ❖ Phase: Lissajous figures (two signals in two deflecting plates)



(Volt/Div : 100mV/Div, Time/Div : 0.5ms/Div)

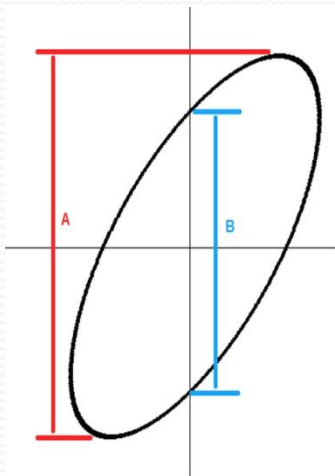
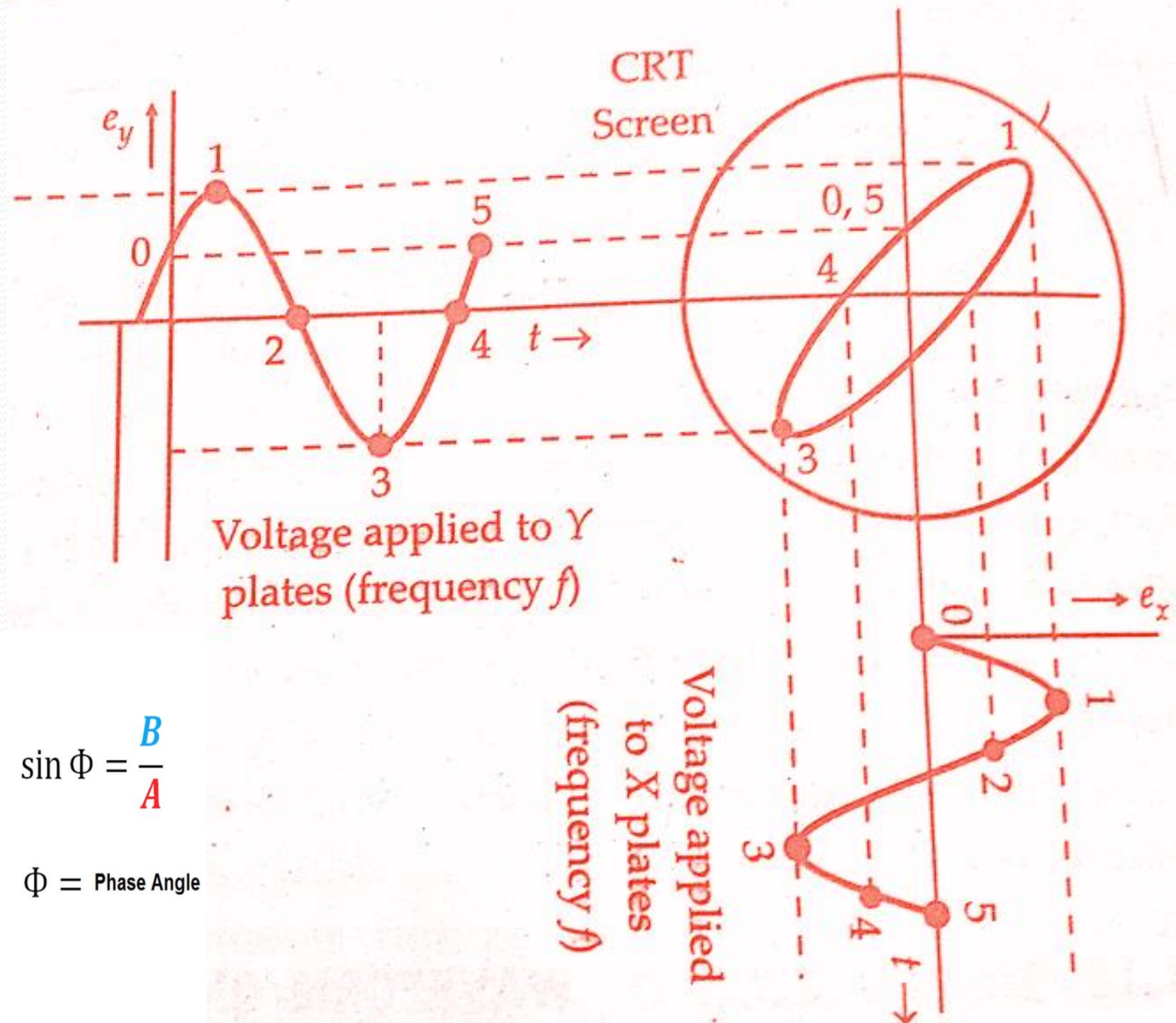
a) Voltage Peak-to-Peak

$$\begin{aligned} V_{p-p} &= (V/Div) \times \text{No. of vert. div.} \\ &= 100 \text{ mV/div} \times (3.8 \times 2) \\ &= \underline{0.76 \text{ V}} \end{aligned}$$

b) Voltage Peak

$$\begin{aligned} V_p &= (V/Div) \times \text{No. of vert. div.} \\ &= 100 \text{ mV/div} \times (3.8) \\ &= \underline{0.38 \text{ V}} \end{aligned}$$

Phase of a signal using Oscilloscope ???



$$\sin \Phi = \frac{B}{A}$$

Φ = Phase Angle